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Where Minds Pollinate

Class 9
Chemistry
Chapter 2
NOTES

NBF book

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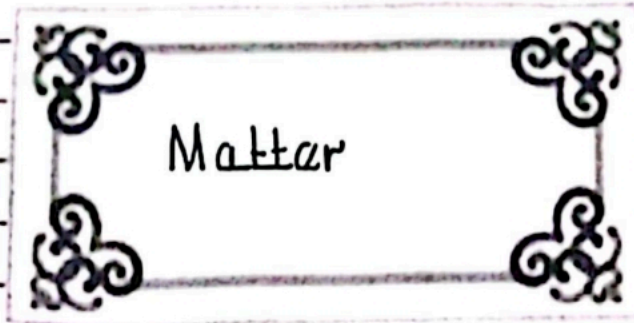


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17th April-24

C.W

Wednesday



Activity:- Questions and Answers

1- Define matter, what are its types? ~~≠~~

A. Matter:-

"Anything which occupy space & has mass is called matter."

Types of matter:-

1- Solid .

2- Liquid .

3- Gas .

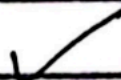
4- Plasma .

2- Explain how energy change matter into different state ? ~~ⓧ~~

A. • Solids become liquids or gases when heated.

• At very high temperature or when subjected to a strong electric field, the gas transforms into plasma.

• Temperature & energy levels on earth are not sufficient to ionize atoms & create plasma.



Activity:-Question & Answers

Q₃: What happen when crystalline Solide are heated?

Ans. Some Crystalline Solids turn into cloudy Liquide called Liquide crystal. Liquid crystals have many properties of Liquids & Some properties of Solids this forms exists within a Certin temperature range when heated more it change to transparent Liquide.

Q₄: What is Bose-Einstein Condensates (BES)?

Ans. It is state of matter in which Separate atoms Goled to temperature very close to absolute zero. Superfluid & Superconductors are two main materials, which contain BES.

Q₅: What are the properties of different states of matters?

Ans. State of matter	Gas	Liquids	Solids
1- Density:-	Low density at normal condition (large space).	High density.	High density.
2- Compressibility:-	very compressible due to large space b/w molecules.	moderately compressible.	Not compressible.
3- Fluidity:-	Can flow.	Can flow.	Can't flow.

What is element, compound & mixture?

Element:

"It is the simplest form of matter made up of single types of atoms."

All elements consist of atoms of same atomic numbers.

Example:

Hydrogen (H_1)

Carbon (C_6)

Nitrogen (N_7)

Compound:

"A substance formed when two or more different atoms chemically combine."

It is completely a new substance, it is completely different from the elements that were used to make it.

Example:

Hydrogen & oxygen gases combine to form water which is a liquid.

Mixture:

"Mixtures are the physical combinations of substances."

A mixture that does not combine contains the same ^{type} particles of particles. The particles would be chemically different from each other.

Example: Tea is an example of mixture, it is made up of milk, water, Tea leave & Sugar all of which have different chemical combination.

Types of mixture:-

Mixture.

a) heterogeneous Mixture.

b) homogeneous mixture.

Q: Define Allotrope & explain with the help of example.

Ans: Allotrope:-

"The property of an element to exist in different physical forms is called Allotrope"

Atoms of some elements arranged in different manners.

in the same physical state in allotropes. They are the different structural forms of the same elements.

Example:-

There are 3 allotrope of carbon

a) Graphite:- Graphite is composed of flat two dimensional layers of hexagonally arranged carbon atoms. In a layers each, C-atom is Covalently bonded to three others carbon atoms. When weak intermolecular force exist between each layer which allows the layer to slide over one another without breaking the bonds. This arrangement makes graphite soft & Slippery, making it ideal to be used as a Lubricant. It is a good conductor of electricity.

Diamond:-

Diamond is the hardest & the purest crystalline allotrope of carbon. In its structure, each C-atom is covalently bonded to four other carbon atoms forming a rigid structure of tetrahedral shape.

It has high melting point, since all the carbon atoms are bonded with other carbon atoms, no free electrons are present, resulting in the structure being non-conductive. Diamond is non-conductors of electricity.

Bucky Balls (C-60):-

Bucky Balls, also known as fullerenes, having a football like fused hollow ring structure made up of twenty hexagons & twelve pentagons. Each of its carbon atoms are bonded to 3 carbon atoms.

Topic:- Solution:-

Q1 Define Solution, what are Solvent & Solute.

Ans Solution:-

A Solution is homogenous mixture of two or more substance in which one substance is dissolved in other.

Solute:-

A substance which is dissolved is called "Solute".

Solvent:-

A substance in which solute is dissolved is called "Solvent".

Example:-

Eg. A sugar/salt dissolved into water.

Q2 What are the type of Solution.

Ans There are three main types of Solution.

- 1- Gaseous Solution.
- 2- Liquid Solution.
- 3- Solid Solution.

Q3 Write a detail note on types of Solution.

Ans Gaseous Solution:-

It can be gas in gas, gas in liquid or gas in solid.

1-) gas in gas:-

Eg: Nitrogen & Hydrogen mixed together in Haber's process to make ammonia gas.

2-) Gas in liquid

Eg:- Fog, cloud & mist are the example of Solution where

Liquid water (Solute) is dissolved in air (Solvent)

Gas in Solid:

Smoke is a solution of Carbon particles in gaseous air. or if we freeze CO₂ gas it combines the air, which is solid through of gas in solid.

Liquid Solution:-

It can be liquid in liquid, liquid in gas & liquid in solid.

a) Liquid in gas:-

Carbonated drinks are solutions where solvent is liquid (water) is solvent & solute is gaseous (carbon dioxide).

b) Liquid in liquid:

Vinegar (acetic acid in water) is an example of liquid dissolved in liquid.

c) Liquid in solid:-

Brine & Sugar Syrup are solution of solid salt and sugar in water.

Solid Solution:-

It can be solid in solid, liquid & gas. ✓

a) Solid in gas:-

Hydrogen gas on nickel metal surface is used in process where hydrogen is solute & nickel catalyst in solvent.

b) Solid in liquid:-

Amalgam solution of any matter (solid) in mercury (liquid)

c) Solid in solid:-

Alloys mixture of metals like, brass, bronze, steel.

Q⁴: Define aqueous Solutions.

A: Definition:- Aqueous Solution is formed by dissolving a substance in water. The dissolved substance may be solid, liquid or gas.

Q⁵: What are true solutions?

In order to be solution, a mixture must be homogeneous.
Examples:- Sugar / table salt in water.

Q⁶: What is universal solvent?

Water is a universal solvent because it dissolves majority of compounds present in earth's crust.

Q⁷: What are saturated, unsaturated & super saturated solutions?

Unsaturated Solution:- A solution that contains lesser amount of solute than that is required to saturate it at a given temperature is called unsaturated solution. Such solutions have the capacity to dissolve more solute.

Saturated Solution:- A solution containing maximum amount of solute at a given temperature is called saturated solution. Further added solute remains undissolved & it settles down at the bottom of container.

Super Saturated Solutions: When Saturated Solutions are heated, they develop further capacity to dissolve more Solute. The Solution is more concentrated ^{than} ~~that~~ ^{than} Saturated Solution. It is called Super Saturated Solution. When such Solution are cooled, Solute crystalline out & leave behind a Saturated Solution.

Q⁸ What are dilute & concentrated Solutions?

Dilute Solutions are those which contain small amount of dissolved Solute. Where as concentrated Solutions are those which contain relatively large amount of Solute in the Solution.

Q⁹ Define Solubility, on which factors does it depend.
Solubility:-

"Solubility is the maximum amount of Solute which dissolved in a specified amount of a Solvent at a specific temperature"

Factors:-

- 1- It depends upon the Solvent used.
- 2- Temperature.
- 3- pressure. ✓

Q¹⁰ What is the effect of temperature on Solubility.

Depending upon the nature of Solute their is either.

• Increase in Solubility with temperature Eg: KCl , NH_4Cl .

• Decrease in Solubility with temperature Eg: Na_2SO_4 , $Ca(OH)_2$.

3rd May-24

C.W

Friday

Q11 Defi Differentiate b/w Solution, Colloids & Suspension

Answer:-

Solution	Colloids	Suspension
* A homogeneous mixture of two or more components.	A heterogeneous mixture of two or more components.	A heterogeneous mixture of two or more components.
* Particle size: vary from 0.1 to 10^3 nm. Not visible by naked eye.	Particle size vary from 1 to 10^3 nm. Visible by naked eye.	Particle size greater than 10^3 nm. Visible by naked eye.
* Particle can pass through as well as Ultra-filter paper.	Particle can pass through normal filter paper but not through ultra filter paper.	Particles can't pass through as well as Ultra-filter paper.
* Cannot Scatter the Light (due to small size)	Can scatter the light (Tyndal effect)	Can scatter the Light (Tyndal effect)
* Do not Separate.	Do not Separate	Separate or settles down.
For example: Water	For example: Milk	For example: Muddy

7th May-24

C.W

Tuesday

Think Tank

Question & Answers

Question no 1:-

Why is Solution Considered mixture? *

Answer:-

A Solution is considered a mixture because the particles of Solute are evenly distributed. Such Solutions are called homogeneous mixtures.

Question no 2:-

How will you test whether the given Solution is a Colloid or a Solution. *

Answer:-

To check whether Solution is a Colloid, pass a beam of light through it. The path becomes visible as a bright streak. This is known as Tyndal effect.

10/5

